EXPERIMENT 5

1. Introduction

This report documents the task analysis and visual user flow for a standard e-commerce application. The goal is to break down complex user interactions into manageable steps to identify potential friction points and optimize the path to conversion (Order Confirmation).

2. Task Analysis Overview

Task analysis was conducted to understand the cognitive and physical steps a user takes to achieve their goal.

- **Goal:** Successfully purchase a product.
- **Key Actions:** Discovery (Search/Browse), Evaluation (Product Details), Selection (Add to Cart), and Transaction (Checkout/Payment).
- **Identified Pain Points:** Potential drop-off during the "Sign In" vs. "Guest Checkout" decision phase and the complexity of entering shipping/payment information.

3. Visual User Flow Analysis

Based on the provided flowcharts, the user journey is structured into a logical sequence with specific decision nodes.

Phase 1: Product Discovery & Selection

- **Entry Point:** Start (Open App).
- **Looping Logic:** The flow includes a critical decision diamond: **"Add to cart?"**.
    - If **No**, the user is redirected back to "Browse/Search products," creating a recursive discovery loop until a selection is made.
    - If **Yes**, the user proceeds to the transactional phase.

Phase 2: Cart & Authentication

- **Review Process:** Users move from "Add to cart" to "Review cart" and then "Checkout."
- **Decision Node: "Have account?"**
    - **Yes:** Redirects to "Sign in."
    - **No:** Redirects to "Guest checkout / Create account."

- **Merging Paths:** Both paths converge at the "Enter shipping info" step to ensure a unified checkout experience regardless of account status.

Phase 3: Finalization

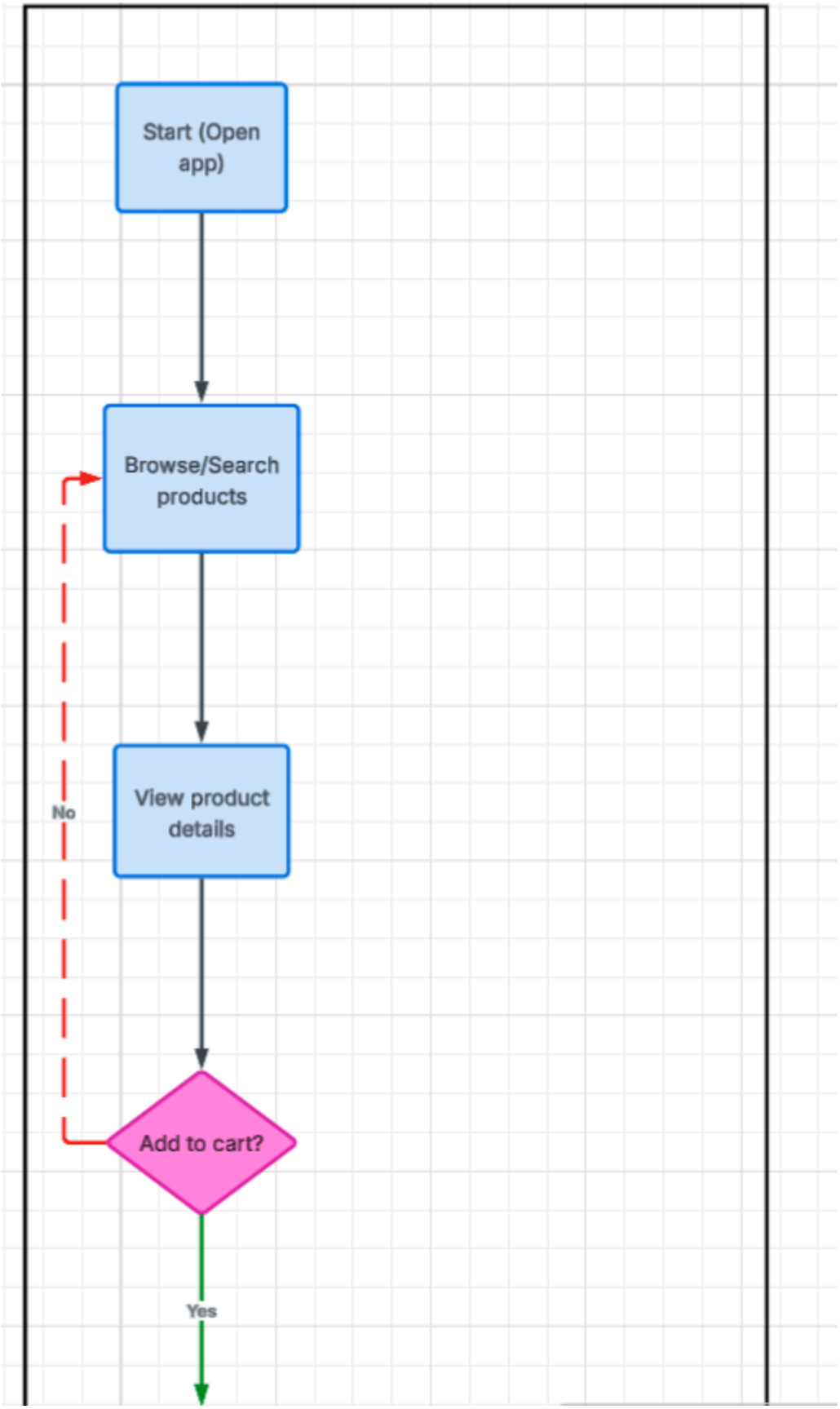
- **Sequential Steps:** Selection of payment method $\rightarrow$ Place order $\rightarrow$ Order confirmation.
- **Terminal State:** The process concludes once the "Order confirmation" is received, satisfying the user's primary goal.

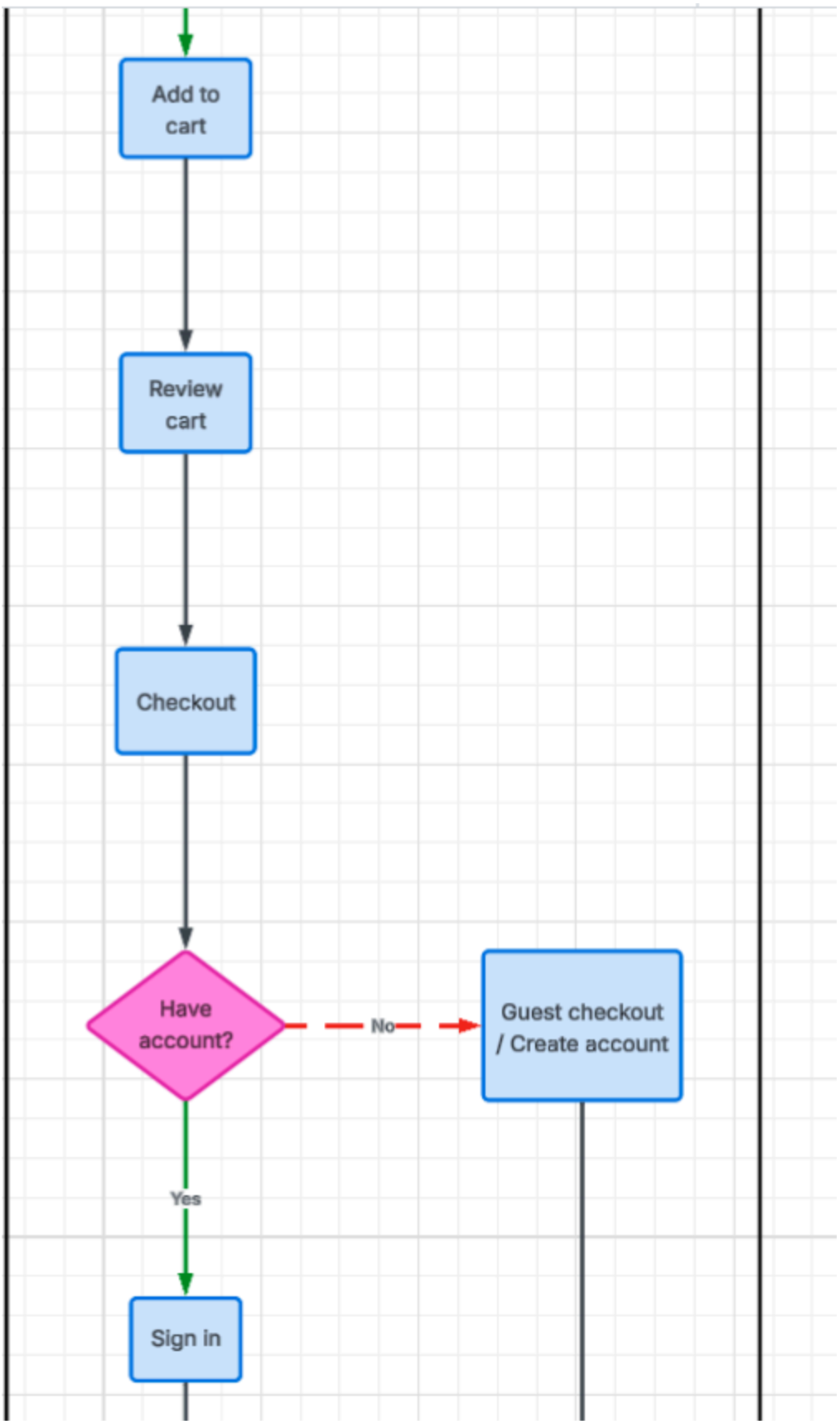
4. Comparison: Conceptual vs. Observed Flow

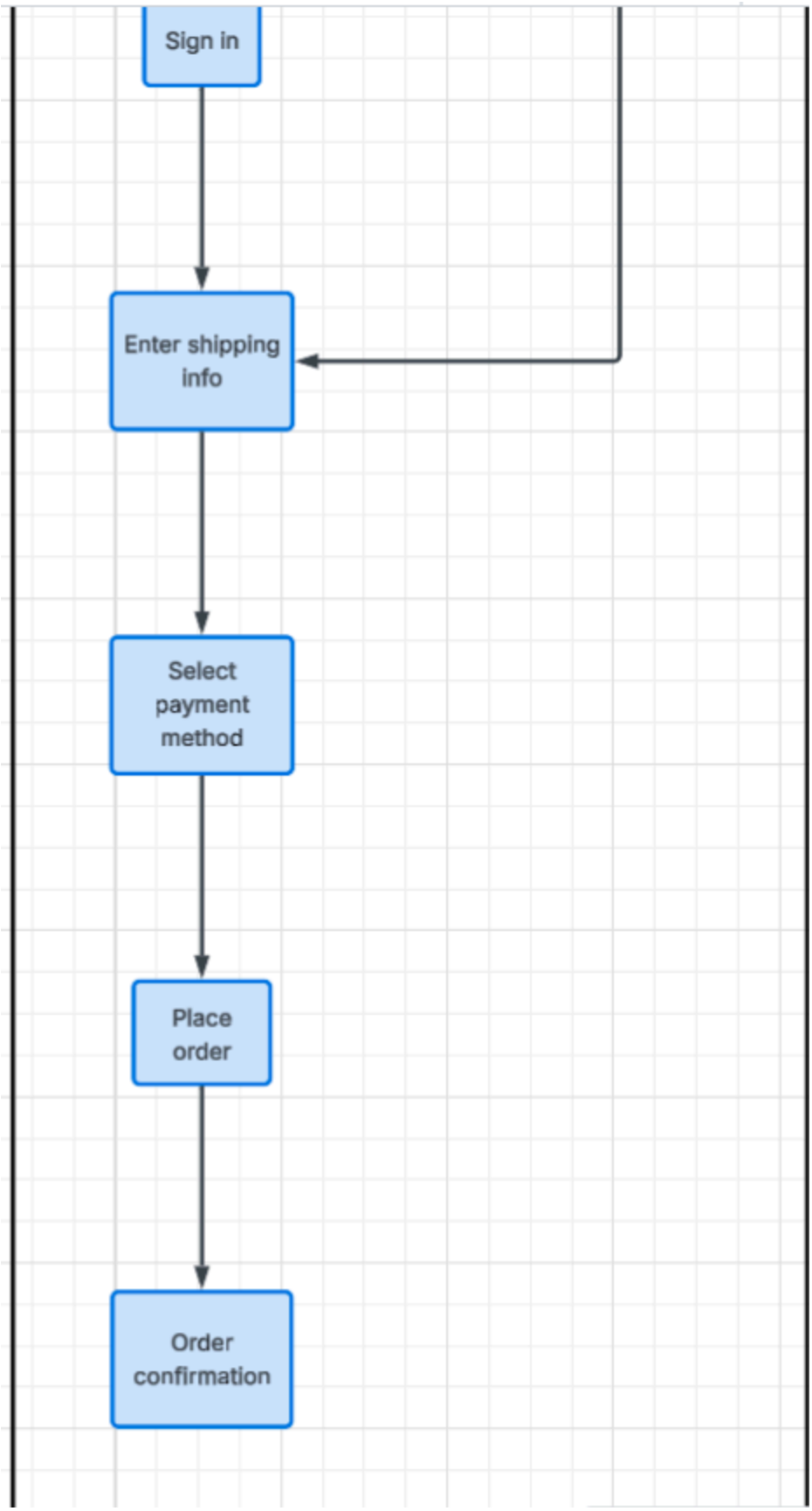
Feature	**Conceptual Text Flow**	**Observed Diagram Flow**
Flexibility	Linear progression.	Includes "No" paths and loops for browsing.
Authentication	Mentions Login/Register at the start.	Places Authentication *after* the cart review (Contextual Login).
Post-Purchase	Includes "Track Order."	Ends at "Order Confirmation."

5. Key Findings & Recommendations

- **Contextual Authentication:** The diagram shows authentication happening at the checkout stage rather than the start. This is a "best practice" as it reduces upfront friction during browsing.
- **Error Handling:** While the "No" path for adding to cart is defined, further design should consider "Error/Invalid Payment" loops during the final phase to prevent user exit.
- **Efficiency:** The flow is streamlined, but the "Enter shipping/payment info" steps are high-effort. Implementing auto-fill or saved profiles for returning users would significantly improve the UX.


Start (Open app)
Browse/Search products
View product details
Add to cart?
No
Yes


Add to cart
Review cart
Checkout
Have account?
No
Guest checkout / Create account
Yes
Sign in


Sign in
Enter shipping info
Select payment method
Place order
Order confirmation